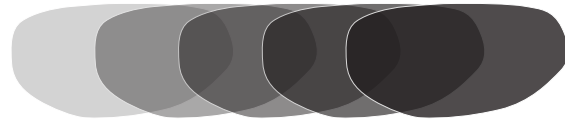


Answer all questions.

1. (a) The diagrams show some uses of smart materials which rely on their unusual properties.

sunglasses



cloudy

sunny

baby bath thermometer



cold



hot

car bumpers



nappies





Examiner
only

The box contains the names of four types of smart material.

thermochromic pigment shape memory polymer

hydrogel photochromic pigment

Use the information given to complete the table. [2]

Unusual property	Use	Type of smart material
it regains its original shape on heating	car bumpers	shape memory polymer
it changes colour with changing temperature		
it can absorb up to 1000 times its volume of water		

3410U201
03



(b) Draw a line to link the use of each nano-material to its property.

[2]

Use

Property

nano-silver in dressings for
cuts and burns

antibacterial

blocks harmful UV light

nano-titanium dioxide in
sunscreen creams

breaks down dirt

strong and light

